



Document ID:	PRJ-2026-004	Version:	1.2	Status:	Final
Author:	Lisa Park, Data Engineer	Print Date:	2026-01-16	INTERNAL USE ONLY	

Document

Purpose

This document defines the database naming conventions, schema standards, and data modeling guidelines for the DataFlow Platform. All database objects must comply with these standards.

Naming Conventions

Tables

- Use snake_case for all table names
- Prefix with domain: `crm\_`, `ecom\_`, `support\_`, `marketing\_`
- Examples: `crm\_customers`, `ecom\_orders`, `support\_tickets`

Columns

- Use snake_case
- Primary keys: `{table\_name}\_id`
- Foreign keys: `fk\_{referenced\_table}\_id`
- Timestamps: `created\_at`, `updated\_at`, `deleted\_at`
- Boolean flags: `is\_` prefix (e.g., `is\_active`, `is\_verified`)

Indexes

- Primary: `pk\_{table\_name}`
- Unique: `uq\_{table\_name}\_{column\_name}`
- Foreign key: `fk\_{table\_name}\_{referenced\_table}`
- Performance: `ix\_{table\_name}\_{column\_names}`

Schema Standards

Required Columns (All Tables)

Every table must include:



Column	Type	Description
id	UUID	Primary key
created_at	TIMESTAMP	Record creation time
updated_at	TIMESTAMP	Last modification time
created_by	VARCHAR(100)	User who created record

Audit Requirements

For tables containing customer PII, additional audit columns are required as specified in security_policy.md, Section 4.2 (Data Audit Trail).

Data Types

Use Case	PostgreSQL Type	Notes
Identifiers	UUID	Always use UUID v4
Money	NUMERIC(15,2)	Never use FLOAT
Timestamps	TIMESTAMPTZ	Always store in UTC
Short text	VARCHAR(n)	Specify max length
Long text	TEXT	For descriptions
JSON data	JSONB	For flexible schemas

Performance Guidelines

Indexing Strategy

- Always index foreign keys
- Create composite indexes for common query patterns
- Use partial indexes for filtered queries
- Review slow query logs weekly

Partitioning

Tables exceeding 10 million rows should be partitioned:

- Time-series data: monthly partitions
- Regional data: by region code

Phase 2 Database Requirements

The following schemas must be ready before Phase 2 launch:

Schema	Purpose	Owner
api_gateway	API rate limiting, tokens	Backend Team



streaming	Real-time event storage	Data Team
audit_log	Security audit trail	InfoSec

Note: The `audit\_log` schema has additional encryption requirements. See security_policy.md for encryption-at-rest specifications.

Migration Standards

- All migrations must be reversible
- Use sequential version numbers: `V001\_\_description.sql`
- Test migrations on staging before production
- Document breaking changes in CHANGELOG

Document Owner: Lisa Park, Data Engineer